

INTERNSHIP REPORT

*A Report Submitted
In Partial Fulfilment
for award of Bachelor of Technology*

**In
COMPUTER SCIENCE AND ENGINEERING (MTech Int.)**

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DECLARATION

I hereby declare that the work presented in this report was carried out by me. I have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute.

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TABLE OF CONTENTS

Declaration	
Acknowledgement	
1. INTRODUCTION	1-2
1.1 ORGANIZATION OVERVIEW	1
1.2 OBJECTIVE AND SCOPE	2
2. INTERNSHIP EXPERIENCE	3-10
2.1 PROJECT WORKED ON	3
2.2 TOOLS AND TECHNOLOGY USED	4
2.3 ROLES AND RESPONSIBILITIES OF THE INTERN	5-6
2.4 DETAILS OF WORK DONE	7-8
2.5 SKILLS LEARNT	9-10
3. CONCLUSION	11-14
REFERENCES APPENDICES INTERNSHIP CERTIFICATES	

INTRODUCTION

1.1 ORGANIZATION OVERVIEW

Infosys Springboard is a digital learning initiative by Infosys Limited, aimed at democratizing access to high-quality education in emerging technologies. It offers free, structured, and industry-aligned learning paths across domains such as Artificial Intelligence, Data Science, Cloud Computing, Full Stack Development, and Cybersecurity, catering to students, professionals, and educators alike.

As part of its skill-development mission, Infosys Springboard conducts periodic internship programs — one of which is the Infosys Springboard Internship 6.0 — that provide learners with the opportunity to apply theoretical knowledge to real-world problems through mentor-guided, project-based learning. These internships are designed to bridge the gap between academic education and industry expectations by exposing interns to practical challenges, structured project timelines, and hands-on tool usage similar to what is expected in a professional AI/ML role.

Under this program, I selected the Artificial Intelligence domain, where I was assigned a healthcare-focused deep learning project. Throughout the internship, I received continuous mentorship that guided me through dataset handling, model development, debugging, and final deployment of the solution.

1.2 OBJECTIVE AND SCOPE

Objective

The primary objective of this internship project was to design, develop, and deploy an AI-based system named CliniScan, capable of analyzing chest X-ray images to assist in the detection of lung abnormalities. Specifically, the project aimed to:

1. Build a multi-label image classification model capable of identifying 14 different thoracic disease categories (plus a "No Finding" class) from chest X-ray images.
2. Implement an object detection model to localize abnormal regions within the X-ray using bounding boxes.
3. Incorporate explainable AI techniques (Grad-CAM) to visualize which regions of the image influenced the model's predictions, thereby improving trust and interpretability for medical professionals.

4. Develop and deploy a user-friendly web application that integrates all the above components into a single, accessible interface.

Scope

The scope of this project extends across several dimensions:

- **Healthcare Assistance:** The system can act as a preliminary screening tool in hospitals and rural clinics where access to expert radiologists may be limited.
- **Educational Utility:** It can be used as a learning aid for medical students to understand common radiological patterns associated with various thoracic diseases.
- **Research Foundation:** The project, built on the publicly available NIH Chest X-ray14 dataset, provides a reproducible foundation that can be extended with more data, additional disease classes, or more advanced architectures in future work.
- **Technology Demonstration:** The project demonstrates an end-to-end AI development pipeline — from data acquisition and preprocessing, through model training and evaluation, to cloud deployment — reflecting real-world software and machine learning engineering practices.

The project does not aim to replace professional medical diagnosis but is intended strictly as a supportive, assistive tool, given the limitations of the dataset size and current model accuracy.

understanding of the software development process, from analyzing requirements and designing application components to implementation, testing, and refinement.

Overall, the scope of the internship was to develop a comprehensive understanding of full stack web development and gain practical experience in building functional web applications by integrating the **frontend, backend, database, and API layers.**

INTERNSHIP EXPERIENCE

2.1 PROJECT WORKED ON

Project Title: CliniScan — Lung Abnormality Detection using AI

During the internship, I was assigned the task of building an end-to-end deep learning solution for chest X-ray analysis. The project was divided into distinct phases:

1. **Data Acquisition and Preprocessing:** Sourcing and preparing the NIH Chest X-ray14 dataset (112,120 images) from Kaggle, including cleaning metadata and converting bounding box annotations into a usable format.
2. **Classification Model Development:** Training a Convolutional Neural Network (EfficientNet-B0) using transfer learning to perform multi-label classification across 15 classes (14 diseases + "No Finding").
3. **Detection Model Development:** Preparing a YOLO-compatible object detection dataset from the NIH bounding box annotations (880 labeled images across 8 disease classes) and training a YOLOv8 model to localize abnormalities.
4. **Explainability Integration:** Implementing Grad-CAM to generate visual heatmaps explaining model predictions.
5. **Application Development and Deployment:** Building a Streamlit-based web application integrating all components, followed by version control using Git/GitHub and deployment on Streamlit Community Cloud.

The final deliverable is a fully functional, publicly accessible web application where a user can upload a chest X-ray image and instantly receive a classification result, a Grad-CAM interpretability heatmap, and a YOLO-based detection output with bounding boxes around detected abnormalities.

2.2 TOOLS AND TECHNOLOGY USED

Programming Language:

- Python

Deep Learning Frameworks and Libraries:

- PyTorch and Torchvision — for building and training the classification model
- Ultralytics YOLOv8 — for object detection model training and inference
- Scikit-learn — for multi-label binarization of disease labels

Model Architectures:

- EfficientNet-B0 (pretrained, fine-tuned via transfer learning) — for classification
- YOLOv8n (Nano variant) — for object detection

Explainability:

- Grad-CAM (Gradient-weighted Class Activation Mapping) — implemented using PyTorch's feature extraction utilities

Data Handling and Processing:

- Pandas and NumPy — for metadata processing and numerical computation
- OpenCV (cv2) and PIL — for image processing and visualization

Web Application Framework:

- Streamlit — for building the interactive web-based user interface

Development and Training Environment:

- Google Colab (with GPU acceleration, NVIDIA Tesla T4)

Dataset Source:

- Kaggle — NIH Chest X-ray14 Dataset

Version Control and Deployment:

- Git and GitHub — for source code and model version control
- Streamlit Community Cloud — for hosting and deploying the final application

2.3 ROLES AND RESPONSIBILITIES OF THE INTERN

As the intern responsible for this project, my key roles and responsibilities included:

1. Requirement Understanding: Studying the assigned problem statement and understanding the structure and challenges of the NIH Chest X-ray14 dataset.
2. Data Engineering: Downloading, cleaning, and preprocessing large-scale medical imaging data; converting raw bounding-box CSV annotations into the YOLO label format required for object detection training.
3. Model Development: Independently implementing, training, and fine-tuning both the classification model (EfficientNet-B0) and the detection model (YOLOv8) using PyTorch and Ultralytics frameworks respectively.
4. Performance Evaluation: Evaluating model performance using appropriate metrics (accuracy, precision, recall, F1-score for classification; [mAP@0.5](#), precision, and recall for detection) and iterating on the approach based on results.
5. Explainability Implementation: Researching and implementing Grad-CAM to make the classification model's predictions interpretable through visual heatmaps.
6. Application Development: Designing and coding a modular Streamlit application (app.py) along with reusable utility modules (gradcam_utils.py, detect_utils.py) following clean project structuring practices.
7. Deployment and DevOps: Managing the project using Git version control, resolving deployment-specific issues (such as missing system dependencies and package conflicts), and successfully deploying the application on Streamlit Community Cloud.
8. Debugging and Troubleshooting: Identifying and resolving multiple real-world technical issues during the project — including dataset path errors, indentation errors, dependency conflicts, and system-level library issues (e.g., OpenCV's dependency on libGL) — with guidance from my mentor.
9. Documentation: Maintaining clear documentation of the workflow, dataset structure, model performance, and deployment steps for future reference and reproducibility.

2.4 DETAILS OF WORK DONE

Phase 1 — Data Preparation: The NIH Chest X-ray14 dataset was downloaded via the Kaggle API, consisting of 112,120 chest X-ray images labeled across 14 disease categories (Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, and Pneumothorax) plus a "No Finding" category. The Finding Labels column was parsed and transformed using MultiLabelBinarizer to support multi-label classification, since a single X-ray can exhibit multiple co-occurring conditions.

For the object detection task, a separate NIH-provided file (BBox_List_2017.csv), containing 984 bounding box annotations across 880 unique images and 8 disease classes, was cleaned and converted into YOLO-compatible label format (normalized x_center, y_center, width, height), including scale correction for image resizing.

Phase 2 — Classification Model Training: An EfficientNet-B0 model, pretrained on ImageNet, was fine-tuned on a 30% subsample of the dataset (approximately 26,900 training images) using a Binary Cross-Entropy with Logits loss function, suitable for multi-label classification. The model was trained for 5 epochs using the Adam optimizer with a learning rate of 1e-4 on a GPU-enabled Google Colab environment. Training loss decreased consistently across epochs (from 0.2176 to 0.1495), and the final model achieved a test accuracy of 92.77%.

Phase 3 — Detection Model Training: A YOLOv8n model was trained on the prepared 880-image bounding-box dataset (704 training images, 176 validation images) for 50 epochs. Given the relatively small dataset size, more training epochs were required to allow the model sufficient exposure to learn meaningful detection patterns. The final model achieved a [mAP@0.5](#) of 0.174 overall, with notably strong performance on the Cardiomegaly class ([mAP@0.5](#) of 0.685), while classes with fewer or more subtle visual patterns (e.g., Effusion, Nodule) showed comparatively lower performance — an expected outcome given the limited annotated data available for detection tasks.

Phase 4 — Explainability (Grad-CAM): To enhance model transparency, Grad-CAM was implemented to generate heatmaps highlighting the image regions most influential to the classification model's decision. This was achieved by extracting intermediate feature maps from the EfficientNet-B0 architecture and computing a weighted activation map, which was then overlaid on

the original X-ray image for visualization.

Phase 5 — Application Development: A Streamlit web application (app.py) was developed to integrate all model components into a single user interface. The application allows users to upload a chest X-ray image and receive, in real time: (1) the top predicted disease classes with confidence scores, (2) a Grad-CAM interpretability heatmap, and (3) a YOLO-based detection output showing bounding boxes with predicted class labels and confidence scores. The project was organized following a clean, modular folder structure (models/, utils/, app.py, requirements.txt) to ensure maintainability and reusability.

Phase 6 — Deployment: The completed project was version-controlled using Git and pushed to a public GitHub repository. It was then deployed using Streamlit Community Cloud. During deployment, several real-world issues were encountered and resolved, including a Python indentation error in the application code and a missing system-level dependency (libGL) required by OpenCV, which was resolved by adding a packages.txt file specifying the correct system package (libgl1) for the deployment environment. The application is now live and publicly accessible.

2.5 SKILLS LEARNT

Technical Skills:

- Practical understanding of Convolutional Neural Networks and transfer learning techniques
- Hands-on experience with multi-label image classification using PyTorch
- Object detection model training and evaluation using YOLOv8 and the Ultralytics framework
- Data preprocessing techniques for medical imaging datasets, including annotation format conversion (CSV to YOLO format)
- Implementation of explainable AI techniques (Grad-CAM) for model interpretability
- Full-stack AI application development using Streamlit
- Version control using Git and collaborative code management via GitHub
- Cloud deployment and troubleshooting of production-level AI applications, including dependency and environment management

Soft Skills:

- Structured problem-solving under real-world constraints such as limited computational resources and small annotated datasets
- Effective time management while balancing model training duration against accuracy requirements
- Debugging and root-cause analysis of technical errors across multiple layers of a software stack (data, model, application, deployment)
- Self-directed learning of new tools, libraries, and platforms not covered in prior academic coursework
- Clear technical documentation and communication of project outcomes

CONCLUSION

This internship, undertaken under the Infosys Springboard Internship 6.0 program in the Artificial Intelligence domain, provided a comprehensive, hands-on introduction to the complete lifecycle of an AI/ML project — from raw data acquisition to a publicly deployed, working application.

Through the development of CliniScan, I gained practical experience in applying deep learning techniques to a real and meaningful problem: assisting in the preliminary screening of thoracic diseases from chest X-ray images. The project involved not only building and training two distinct deep learning models — a multi-label classifier (EfficientNet-B0) achieving 92.77% accuracy and an object detector (YOLOv8) achieving a [mAP@0.5](#) of 0.174 — but also integrating an explainability layer (Grad-CAM) to make model predictions more transparent and trustworthy, a crucial requirement in any healthcare-related AI application.

Beyond the technical modeling work, this internship offered valuable exposure to the practical, often-overlooked aspects of real-world AI development: managing large datasets under limited computational resources, structuring a project for maintainability, using version control effectively, and diagnosing and resolving deployment-specific issues such as environment and dependency conflicts. These challenges, while sometimes difficult, significantly strengthened my problem-solving abilities and technical resilience.

The guidance and mentorship provided throughout the Infosys Springboard program played a pivotal role in helping me navigate these challenges and successfully deliver a complete, functioning product. I believe the skills gained — spanning deep learning, model evaluation, explainable AI, and deployment engineering — will be directly valuable in my future academic and professional pursuits in the field of Artificial Intelligence and Machine Learning.

While the current version of CliniScan achieves reasonable results given its constraints (a relatively small annotated dataset for detection, and limited training epochs for classification), there remains significant scope for future improvement, including training on the full dataset, incorporating additional disease classes, experimenting with more advanced architectures, and potentially validating the system with real clinical feedback. Overall, this internship has been an enriching and transformative learning experience that has deepened both my technical expertise and my appreciation for the end-to-end discipline required to build responsible, real-world AI systems.

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OFFER LETTER

Infosys Springboard Virtual Internship 6.0 | Agile Methodology

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INTERNSHIP CERTIFICATE



CERTIFICATE OF COMPLETION

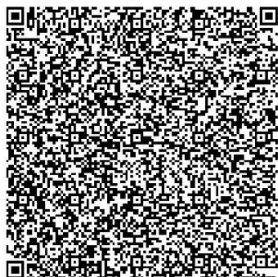
presented to

Pratigya ---

for actively participating and completing the mandatory assignment related to

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